What Simulation Predicts About a Shifting Metagame: A Study in Pokémon and Hearthstone

Alex Elton-Pym and Kazjon Grace

***Abstract*—Balance patches and expansions shift a competitive game's metagame, the mix of cards or species its players favor. We test what simulation actually predicts about that shift, in two games with independent pipelines: a replication of Saravanan and Guzdial's meta discovery framework in Pokémon Showdown, extended from four hand-picked bans to 42 consecutive months of ordinary drift, and a marginal-value probe driving evolutionary deckbuilding in Hearthstone, predicting card adoption across the Naxxramas expansion and the Starving Buzzard nerf against 33,697 real decklists. The two studies agree: a no-change baseline is hard to beat at naming the metagame's members, since most of a metagame is stable; what simulation does predict is how much the metagame will move; methods that feed on their own predictions degrade until grounded in real data; and the real usage a pipeline starts from carries more predictive content than the simulation on top of it. We contribute two methods aimed at the movement signal and an evaluation standard for future work on this task.**

***Index Terms*—Metagame, game balance, replication, Pokémon, Hearthstone, evolutionary computation.**

I. Introduction

DEVELOPERS of competitive multiplayer games adjust balance continuously, and each adjustment can move the metagame, the set of characters, cards, or strategies the player base currently favors. Designers would like to anticipate that movement before a change goes live, and a line of work on automated balancing evaluates proposed changes inside a simulator. For most of its history that work scored its predictions against a metagame internal to the simulator, with no check against what a real player population went on to do. Saravanan and Guzdial [1] closed that gap for Pokémon Showdown: their meta discovery framework builds teams from usage statistics, plays them against each other with an automated agent, and reads a predicted metagame off the resulting pick statistics, validated against Smogon's measured post-ban usage for four historical bans, where it recovered 92.5–100% of the true post-ban meta.

Four hand-selected ban scenarios establish that the method can work. They leave open how wide its operating envelope is, how it behaves when the balance change is small or absent, and which part of the pipeline carries the predictive content. A single-game demonstration also leaves open whether any conclusion is a property of metagame prediction or a property of Pokémon. We take up both questions with two studies in two games, run as independent pipelines with independent codebases and ground truth.

The authors are with The University of Sydney, Sydney, Australia (e-mail: alex.elton-pym@sydney.edu.au; kazjon.grace@sydney.edu.au).

The first study stays in Pokémon Showdown and treats the original framework directly. We reimplement its best configuration and evaluate it on a completed $\{\text{generation}\}\times\{\text{rating cutoff}\}$ grid: the four original Generation 8 bans rerun at the full battle budget, and 42 consecutive monthly transitions of Generation 9 OverUsed (gen9ou), most of which contain no balance change and therefore stress a method built for balance changes. Monthly usage files are published on a fixed schedule in a fixed format, so the drift regime supplies dozens of objective test cases where the original had four.

The second study moves to Hearthstone, a collectible card game, and builds a different pipeline for a different shock type. Where a ban removes a legal option, an expansion adds new cards with no prior usage (thirty, for Naxxramas) and a nerf leaves a card legal at a lower power level. We seed evolutionary deckbuilding with the real pre-shock decks, played by self-play-trained search agents on a purpose-built engine, and, because whole-deck evaluation cannot resolve the value of a single card at feasible budgets, we bias the search with a marginal-value probe that measures each card's contribution in fixed real host decks. Ground truth is 33,697 ranked 2014 decklists spanning the Naxxramas expansion and the September 2014 Starving Buzzard nerf.

Read together, the two studies agree on more than either shows alone:

1) *Composition is the wrong yardstick.* In both games a no-change baseline wins the standard composition-overlap metric: Pokémon no-change beats the framework on overlap in every cell of the grid, and the Hearthstone no-change baseline ties or beats the pipeline on the overall adoption rank correlation. Overlap saturates near its ceiling because most of a metagame is stable month to month, so it measures stability more than prediction.
2) *Movement is the signal a simulator recovers.* What both pipelines predict is how much the metagame moves. Pokémon self-play matches the magnitude of monthly ranking change better than no-change in 42 of 42 transitions at a high-ladder cutoff, and Hearthstone probe-biased evolution predicts the direction and rough size of adoption changes on the cards that actually moved, including the Buzzard collapse from 73% to a predicted 6% against a real 17%.
3) *Self-referential usage loops fail the same way in both.* The Pokémon framework's usage refresh feeds the team

builder its own output, and the Hearthstone free-running forecast feeds evolution its own previous population as the evaluation field. Both drift from what players did: battle outcomes do not enter the Pokémon usage update, while Hearthstone's simulated selection does not prevent drift. In both, grounding the method in real data improves the forecasts.

4) *Use of simulated outcomes.* The tested Pokémon ABC-Meta configuration builds teams from real usage and base stats, then predicts usage by counting appearances on those teams. Wins and losses do not affect the counts, so removing the battles leaves the predictions almost unchanged. In Hearthstone, simulated games measure the win-rate change from inserting a card into a real deck. These measurements guide mutation, and further games determine which evolved decks survive. Unguided evolution did not reliably find the relevant card improvements at the budgets tested. Our Pokémon surrogate learns to estimate matchup win probabilities from simulated games and uses those estimates to select teams. Its usage forecast is then blended with the previous month's observed usage.
5) *Two methods that target movement.* For Pokémon we contribute a fast learned stand-in for the battle simulator whose predictions, read out by blending back toward last month's meta, track how much the meta moves two to three times more closely than assuming no change, at about one-seventeenth of simulation's cost. For Hearthstone we contribute the marginal-value probe and probe-biased evolution, piloted by self-play agents under hidden information, with an ablation showing that the information the pilots have changes which cards the search finds.

We regard the Pokémon study as a friendly replication: the original results reproduce on their own scenarios, and the cells where the framework loses pair the original's data with regimes it never evaluated or claimed. The Hearthstone study is a new pipeline that reaches the same methodological conclusions by an independent route. None of these findings depends on the other game, so we can state them as properties of the task rather than of one title.

II. Related Work

Automated analysis of game balance spans several genres. Jaffe et al. [2] proposed estimating competitive balance from repeated games between AI agents, Volz et al. [3] balanced a card game on simulated play alone, and Tomašev et al. [4] used AlphaZero to evaluate alternative chess rule sets. de Mesentier Silva et al. [5] evolved balance changes to Hearthstone cards toward a target win rate, and Hernandez et al. [6] balanced competitive games by optimizing over a parameterized metagame graph; Kavanagh et al. [7] map an evolving metagame across rule configurations by model checking, for games small enough to enumerate, and Pfau et al. [8] balance by simulating agents cloned from player behaviour. These search for a change. A second line measures or predicts what players do after one, without simulating the game: Yue et al. [9] predict which Dota 2 heroes the next patch will change from match records, He et al. [10] show that League of Legends patches affect players unequally by skill, Lee and Ramler [11], [12] track how champion usage follows content updates and find strategies that beat the settled meta, Elbert et al. [13] measure several layers of adaptation to CS:GO patches, Pedrassoli Chitayat et al. [14] show that esports analytics models break at each patch, Heo and Kim [15] predict strategy switches in Clash Royale, whose meta is reshaped by frequent balance updates, and Codsi and Vetta [16] fit a dynamic metagame model to real Smash Bros. character selection. Saravanan and Guzdial [1] instead predict the player-base response to a change by simulation and validate it against real usage; their framework is the direct subject of our first study and the methodological anchor of the second.

Pokémon Showdown [17] has been a testbed since the Showdown AI competition [18]; Huang and Lee [19] trained self-play battlers for it, the PokeAgent Challenge [20] now runs Showdown self-play at scale, and the poke-env library [21] provides the agent interface used in the original study and here. Metamon [22] trains competitive Showdown agents at scale; we use battles from Metamon-derived agents as training data for our surrogate. The VGC AI competition frames Pokémon team building as a meta-game balance problem [23], [24], and Reis et al. [25] pit team-building agents against a balancing agent that rewrites Pokémon attributes toward a diversity target; that line changes the game to balance its meta, where we hold the change fixed and predict the response. Ramos and Soria [26] analyse the Pokémon trading-card metagame against tournament usage through its Nash equilibria. For Hearthstone, the game has served as an AI testbed for a decade. García-Sánchez et al. [27] established evolutionary deckbuilding, Bhatt et al. [28] explored the Hearthstone deck space with it, Fontaine et al. [29] mapped that space with quality-diversity search, Zhang et al. [30] replaced the simulator inside that search with a learned surrogate, and França et al. [31] evolve variants of real decks around a retained core; Q-DeckRec [32] builds a deck against one named opponent by reinforcement learning. Closest to our card probe, Chen and Guy [33] score generated cards inside an evolved meta and find that a card's strength depends on the meta it is scored against, and Long et al. [34] predict the point cost a wargame unit should carry from play. The Hearthstone AI competition [35] drove play agents, including MCTS with learned components [36]. Justesen et al. [37] showed that in multi-action adversarial games the effective unit of decision is the turn plan. Determinized search has a long record in hidden-information card games [38], and Dockhorn et al. [39] found that a Hearthstone search agent gains substantially from knowing the opponent's hand and that predicting the hand recovers most of that gain; our agents play under hidden information throughout, and Section V-F measures directly what visible hands would change. Community analytics, the Vicious Syndicate Data Reaper reports [40], Smogon usage statistics [41], and the HearthPwn deck archive [42], supply ground truth. Both studies draw on common machinery: MAP-Elites [43] for search and Deep Sets [44] for permutation-

invariant team and deck encodings.

III. A Common Problem, in Three Questions

Both studies ask the same thing. A game's *metagame* is the mix of cards or species players actually run [5], [45], which we summarise as a ranking by how often each is used. Let A be that ranking just before a rules change (the pre-shock metagame), B the ranking just after (the post-shock metagame), and B' a method's prediction of B from A and the rules change alone. We judge B' by three questions.

- **Which units?** Did B' name the right cards or species, the ones that make up the bulk of the meta? We call this *composition*: overlap of the top-k lists of B and B' for Pokémon, and a rank correlation between B' and B over all cards for Hearthstone.
- **How much did it move?** A meta drifts a little most months and lurches after a big change. Did B' move about as far from A as B really did, rather than sitting still or overshooting? We call this *movement*: for Pokémon, the edit-distance gap $|ED(A,B) - ED(A,B')|$, where ED is the mean rank shift over shared units; for Hearthstone, the average error in predicted adoption.
- **Which way?** Of the units that genuinely moved, did B' call the rises and falls correctly? We call this *direction*: the fraction of real movers whose direction it got right; for Pokémon we also report Spearman's ρ between the real rank changes $A \to B$ and the predicted $A \to B'$.

Every method is measured against the same dumb yardstick: *no change*, the guess $B' = A$ (the naive forecast of the forecasting literature [46]), with a banned unit removed where a ban applies. No change is deceptively hard to beat on the first question and trivial to beat on the third. Most of a meta is stable across a change, so copying this month forward already names most of next month's units, and a method that perturbs the meta can only lose ground there. But no change predicts that nothing moves, so it is wrong on every mover and scores zero on the third question by definition. A prediction earns its keep on the second and third questions, how much the meta moves and which way, not on the stable bulk the first rewards.

IV. Study 1: Pokémon Showdown

A. The Original Framework

The framework [1] has a battle agent, a team builder, and a simulation environment that runs battles and compiles pick statistics. Its ABC-Meta task assumes the true pre-change metagame and builds six-species teams sequentially, scoring each species by

$$\mathrm{Score}_{\mathrm{ABC}}(x) = \mathrm{Pickrate}(x) \times \mathrm{BST}(x), \tag{1}$$

where $\mathrm{Pickrate}(x)$ is the fraction of team slots holding species x in current usage and $\mathrm{BST}(x)$ its normalized base stat total, a form selected by a grid search over candidate components. Selection is ϵ-greedy ($\epsilon = 0.001$), movesets come from Smogon's per-species statistics, and the environment simulates 450,000 battles per scenario, regenerating teams and refreshing the tracked usage every 1,000 battles. The predicted metagame is the top-40 species by resulting pickrate. On four Generation 8 single-ban scenarios (OU, UU, NU, PU) the method reached overlap 0.925–1.000 against a ban-aware naive baseline's 0.925–0.975, an edge of one species in three scenarios and a tie in the fourth; edit-distance delta 0.11–2.74 against 3.04–6.84; and Spearman's ρ from -0.15 to 0.19, never significant. The authors state that their discovered meta reproduces composition and amount of change but not the specific movement, and that the naive baseline has high overlap. Both caveats organize what follows.

B. Method

We reimplemented the ABC-Meta pipeline from the paper's description, using poke-env and Pokémon Showdown as the original did. Figure 1 shows the three pipelines we compare. The team builder draws species by (1) from the input month's real usage, samples a full set per species from that month's moveset statistics, and validates every team with the @pkmn/sim validator [47]. For each transition we generate two pools of 300 valid teams and play 10,000 self-play battles between two `SimpleHeuristicsPlayer` instances on a local server, reading predicted pickrate off appearances in finished battles. The exact-scenario rerun (Section IV-H) instead uses the full 450,000-battle budget, the refresh schedule, the banned species removed from the pool, and the Little Cup blanket ban, matching the original.

The drift evaluation deviates from the original in ways we record so readers can attribute each result: 10,000 battles rather than 450,000; frozen month-M statistics in the headline arm, with a faithful refreshing variant in Section IV-F; the heuristic agent only, which the original found statistically indistinguishable from its RL agent; single-month Generation 9 transitions rather than three-month Generation 8 ban windows; both rating cutoffs, the high-ladder 1695 files and the unfiltered all-ratings files that Section IV-H ties to the original's own data; and top-20 lists in the drift arm, where a 40-species list would cover most of the smaller monthly universe. The gen8 rerun removes every deviation but the agent choice.

C. Movement Magnitude by Rating Cutoff

At the high-ladder cutoff the framework's strongest claim holds. ABC-Meta's edit-distance delta averages 3.87 on the full roster against no-change's 21.21 (Table I), and it is closer to the true drift magnitude in 42 of 42 transitions (Wilcoxon $p \approx 4.5 \times 10^{-13}$). Since no-change's delta is the drift itself, the simulation reproduces roughly the right amount of ranking movement where predicting none misses by the full drift. This survives a 45× smaller battle budget in a new generation, with no ban to anchor the change.

At the all-ratings cutoff, the one Section IV-H identifies as the original's own data, the same comparison reverses. ABC-Meta's mean delta is 24.09 against no-change's 14.54, closer in only 5 of 42 transitions ($p = 1.9 \times 10^{-4}$). The mechanism is in the two no-change rows of Table I: the all-games meta drifts less than the high-ladder meta on the same months (14.54 against 21.21 rank positions), because casual players spread usage widely and shift slowly. The simulated meta has

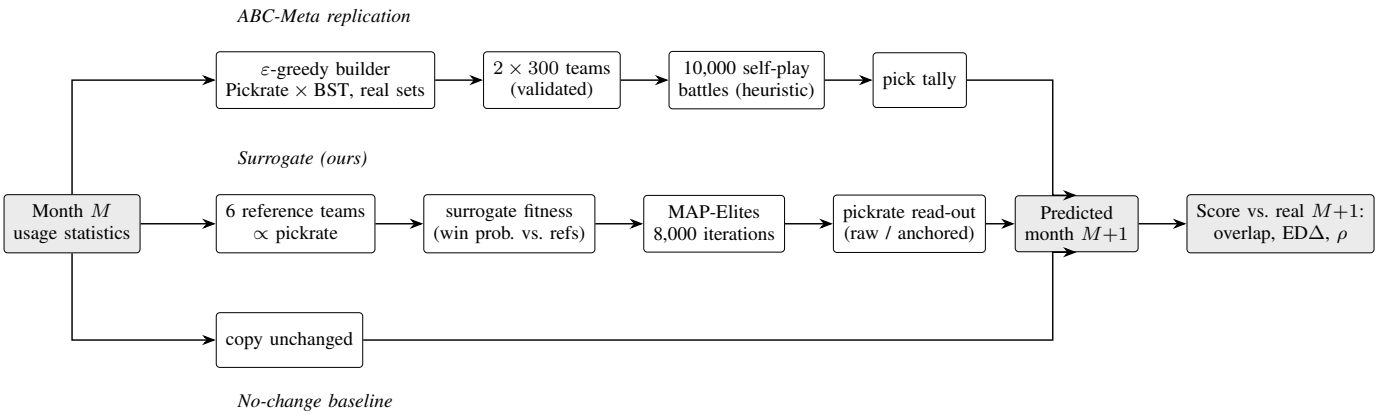


Fig. 1. The three Pokémon prediction pipelines. All start from the same real month-M usage statistics and are scored against real month $M+1$ with the same three metrics.

the opposite character, because the team builder, scoring each species by its usage times its stat total, piles onto a few high-scoring species. Aimed at the sharp high-ladder target it moves by about the right amount; aimed at the flat all-games target the same output implies two to three times the movement that occurs. Nothing here contradicts a published number. The failing cell pairs the original's cutoff with a regime it never evaluated.

D. Composition Results

No-change reaches 0.886 mean top-20 overlap on the full roster and 0.907 in-vocabulary; ABC-Meta reaches 0.833 and 0.875, losing the paired comparison 31–2 ($p = 2.6\times10^{-6}$) and 25–3 ($p = 6.7\times10^{-5}$). When 17 or 18 of next month's top 20 are this month's, any predictor that reproduces the current meta scores high overlap, and a simulation that perturbs the meta can only lose ground unless its perturbations are unusually well aimed. Overlap here sits near its ceiling for reasons unrelated to prediction, so it measures metagame stability more than predictor skill. This does not indicate an error in the original study, whose ban scenarios forced real compositional change and gave overlap something to detect. It means the overlap advantage is specific to that setting and should not be expected in the routine months that make up the large majority.

E. The Battle-Free Ablation

Three independent observations show that in the ABC-Meta configuration the battles' outcomes have no path into the prediction. First, (1) contains no outcome term; the tracked pickrates are updated only from the teams the builder generated, so a battle influences the statistics only by finishing and contributing its teams to the counts. Second, the original's own tables report identical overlap for its RL and heuristic agents (GXE 42.4% against 49.4%) in all four scenarios, which is what outcome-independence predicts. Third, the pipeline runs with the simulator deleted: the same builder, refresh schedule, and 450 updates with zero battles reproduce the full-budget predictions almost unchanged (Table II, No-battle columns), in minutes on one core. The 8–14 hours of simulation per scenario buy sampling precision over the builder's output distribution and nothing else. BSD-Meta, the framework's other task, does consume winrates through its score, so this concerns only the ABC configuration, whose grid-search winner happened to exclude every outcome-dependent term.

F. The Usage-Refresh Ablation

Our headline arm freezes the month-M statistics. Implementing the original's refresh faithfully, recomputing tracked pickrates from accumulated picks every 1,000 battles, costs accuracy: mean overlap falls from 0.833 to 0.801 (paired 25–5, $p = 0.0009$) and edit-distance delta worsens from 3.87 to 5.82 ($p = 5.0 \times 10^{-5}$). The original authors described the mechanism themselves, warning that scoring on a statistic the builder updates would create an "undesirable feedback loop" [1] that concentrates picks. ABC-Meta contains that loop: its score reads pickrates, its refresh rewrites pickrates from the builder's own output, and no battle outcome enters to correct it. Freezing the statistics removes an uncorrected positive feedback, so the real month's data grounds the whole run. The refreshing variant still beats no-change on movement in 42 of 42 transitions, so the central result stands under both configurations; the refresh only erodes the margin.

G. The Surrogate Arm

The replication's cost is dominated by simulation, so we ask how much survives when a learned model replaces it, the surrogate-assisted search of evolutionary computation [30], [48]. A Deep Sets encoder [44] embeds each team as a permutation-invariant set, and a skew-symmetric bilinear head scores a matchup so that $P(\text{A beats B}) = 1 - P(\text{B beats A})$ by construction. A 430-species model trained on 120,000 heuristic self-play battles reaches held-out strength correlation 0.740 and trains in under five minutes on a consumer CPU. To predict a transition we run MAP-Elites [43] for 8,000 iterations over a 12×12 archive of speed and bulk, with fitness the surrogate's win probability against a reference field sampled from month-M usage.

The search leaves us with a pool of strong, varied teams, and turning that pool into a predicted usage ranking is a modelling choice that makes or breaks the result. The obvious way,

TABLE I
POKÉMON MAIN RESULTS OVER 42 MONTHLY TRANSITIONS (MEAN ± SD).

Method	Universe	Overlap (top-20)	Edit-distance Δ	Spearman ρ
ABC-Meta (replication)	full roster	0.833 ± 0.119	3.87 ± 5.28	0.025 ± 0.108
ABC-Meta (replication)	in-vocab	0.875 ± 0.095	1.84 ± 2.21	0.065 ± 0.144
Surrogate, raw read-out (ours)	in-vocab	0.293 ± 0.069	23.17 ± 4.80	0.078 ± 0.117
No-change	full roster	0.886 ± 0.118	21.21 ± 5.13	undefined
No-change	in-vocab	0.907 ± 0.089	5.00 ± 2.23	undefined
ABC-Meta (replication)	full roster, rating 0	0.815 ± 0.117	24.09 ± 7.22	0.019 ± 0.106
No-change	full roster, rating 0	0.888 ± 0.110	14.54 ± 9.60	undefined

Overlap and ρ: higher is better; edit-distance delta: lower is better. Unlabeled rows are scored against the 1695 high-ladder statistics; rating-0 rows are the all-ratings rerun of Section IV-C. No-change has no defined ρ, and its edit-distance delta equals the true drift magnitude by construction.

counting how often each species shows up among the strong teams, predicts far too much upheaval: it names the right units only 0.29 of the time against no change's 0.91. What works is to let the strong teams play each other out, keep the ones that survive that internal competition, and then *blend the result back toward last month's meta* (the same no-change yardstick from Section III). The blend supplies the stable bulk; the model supplies the movement on top. With its settings fixed on the first 21 months and tested on the other 21, this read-out tracks how much the meta moves two to three times more closely than no change, at about 45 seconds per month against the full simulation's 13 minutes, and it keeps that edge on the harder all-games data where the full simulation loses. Two honest caveats. The winning read-out was chosen after seeing early results, so a test on genuinely future months is still owed. And a control that replaced the model's output with random noise, blended the same way, did not reproduce the result, so the model is doing real work rather than the blend alone; what it does not do is call which way individual species move any better than anything else here.

H. Replicating the Four Original Scenarios

To separate what fails to transfer from what fails to reproduce, we reran the four Generation 8 bans at the full 450,000-battle budget with the refresh schedule intact. The unfiltered all-ratings files closely reproduce the paper's published naive baselines (identical top-40 overlap in OU, UU, NU, one species apart in PU), and their appendix's pickrate-only grid-search row sits on top of our all-ratings OU baseline, so we take the all-ratings files as the original's data. Table II reports the rerun: overlap within one species of the paper in every tier, matching outright in PU, and the sign of Spearman's ρ matching in all four scenarios, a nontrivial agreement given that the original found none significant. The overlap edge over the naive baseline is one species in PU and a tie elsewhere, mirroring the original's one-species-or-tie margins and confirming that margins this small need paired tests over many cases. At high-ladder cutoffs (bottom block) the framework never wins overlap and still beats the baseline on movement in three tiers of four, matching the drift arm's split, and the battle-free pipeline reproduces the live run at both cutoffs.

I. The Pokémon Grid

Table III compresses the study. The framework never wins composition in any cell; its best showing is a single species in one ban scenario. Movement carries all of its value, and that value is conditional: simulation tracks how much a concentrated or shocked meta will move, and on flat all-games ordinary drift it loses even that. Which way individual species move stays unpredicted everywhere. Of the species whose real usage shifted by more than five points between adjacent months, ABC-Meta called the direction, rise or fall, right about half the time (0.53). No change is wrong on every mover by construction, so it sets no bar here; direction is a question neither method answers.

V. STUDY 2: HEARTHSTONE

The second study asks the same question in a collectible card game, with an independent codebase and ground truth, where the shocks are an expansion and a nerf rather than a ban.

A. Simulator Fidelity

The simulator is a purpose-built engine for Hearthstone's 2014 Classic era. It implements the full 2014 pool (382 collectible Classic cards and the 30 Naxxramas cards) on the era's rules, and the 245 cards the three classes in this study (Hunter, Mage, Warrior) can play are each covered by behavioural tests. A patch hook applies the September 2014 balance changes (Starving Buzzard 2-mana 2/1 to 5-mana 3/2; Leeroy Jenkins 4 to 5 mana), giving three balance states: the pre-expansion pool, the expansion at launch, and the post-nerf pool.

Decks are piloted by a search agent. A policy and value network reads the game as a set of entities (each minion, hand card, and secret is a token attended over by a single attention layer) and guides PUCT search [49], [50], trained by self-play [51] separately for each balance state, so the post-nerf agent has learned to play the nerfed cards. All experiments use 200 search iterations per move. The agents are trained and run under hidden information: a pilot sees its own hand and both boards, and never the opponent's hand. A matching set of agents trained with both hands visible serves the ablation in Section V-F.

Against a community matchup matrix from roughly 90k tracked games in the 2021 Classic format, which replays the

TABLE II
THE FOUR ORIGINAL POKÉMON SCENARIOS RERUN AT THE FULL 450,000-BATTLE BUDGET, TOP-40 LISTS. PAPER VALUES ARE THE HEURISTIC-AGENT COLUMNS OF THE ORIGINAL'S TABLES II–IV; NAIVE COLUMNS DELETE THE BANNED SPECIES FROM PRE-BAN USAGE. NO-BATTLE IS THE IDENTICAL PIPELINE WITH THE SIMULATOR REMOVED (SECTION IV-E). TOP BLOCK: THE ORIGINAL'S ALL-RATINGS STATISTICS. BOTTOM BLOCK (HL): HIGH-LADDER STATISTICS (1695 FOR OU, 1630 OTHERWISE), WHERE THE ORIGINAL REPORTS NO VALUES.

	Overlap					Edit-distance Δ					Spearman ρ		
Scenario	Ours	No-batt.	Naive	Paper	P. naive	Ours	No-batt.	Naive	Paper	P. naive	Ours	No-batt.	Paper
OU	0.950	0.950	0.950	0.975	0.950	2.65	2.70	2.75	1.09	3.04	$+0.06$	$+0.06$	$+0.18$
UU	0.975	0.975	0.975	1.000	0.975	3.38	3.43	4.82	1.50	4.60	-0.19	-0.20	-0.13
NU	0.925	0.925	0.925	0.950	0.925	4.12	4.01	5.46	2.49	5.57	-0.27	-0.13	-0.07
PU	0.925	0.950	0.900	0.925	0.925	1.36	1.87	5.06	0.11	6.84	$+0.19$	$+0.14$	$+0.15$
OU (HL)	0.900	0.900	0.900	—	—	3.64	3.64	3.64	—	—	$+0.14$	$+0.21$	—
UU (HL)	0.925	0.925	0.950	—	—	4.16	4.00	5.39	—	—	$+0.14$	$+0.11$	—
NU (HL)	0.825	0.825	0.825	—	—	5.19	4.83	6.27	—	—	-0.06	-0.02	—
PU (HL)	0.900	0.900	0.900	—	—	3.80	3.60	7.44	—	—	-0.06	-0.08	—

TABLE III
THE COMPLETED POKÉMON {GENERATION}×{CUTOFF} GRID. EACH CELL ASKS WHETHER THE FRAMEWORK BEAT ITS NO-CHANGE BASELINE PER TRANSITION OR PER SCENARIO; TIES ARE OMITTED FROM W–L COUNTS.

Cell	Trigger	Movement (ED Δ)	Composition
gen9, 1695	drift	won 42–0	lost 2–31
gen9, all games	drift	lost 5–37	lost 3–32
gen8, all games	4 bans	won 4/4	1 win, 3 ties
gen8, high ladder	4 bans	won 3/4, 1 tie	1 loss, 3 ties

2014 card pool at its June 2014 balance [40], the agent's simulated matchup win rates have a mean absolute error of 3.0 pp under hidden information and 2.8 pp with visible hands, at 100 games per matchup cell, a statistical tie. We report absolute error rather than rank correlation because both the real and the simulated matrices sit within a few points of 50%, where rank order is mostly noise.

B. Ground-Truth Data

Ground truth comes from the HearthPwn deck archive [42], [52]: of its 560,660 ranked-mode decks, the 33,697 dated 2014 form our corpus, normalized to the three-class world. One correction matters: deck dates are creation dates but card lists reflect the last edit, so a June deck edited in August can carry Naxxramas cards before Naxxramas existed; filtering on each card's expansion tag removed 3,725 such decks from the pre-expansion window. The nerf produced a dissociation the pipeline must respect. Starving Buzzard's within-Hunter adoption falls from a 67.7% pre-nerf mean to 16.9% afterward (the monthly series; the tag-filtered windows used elsewhere run higher, 73.4% in the evolution seeds and 88.2% in the first period of Fig. 3, because later-edited decks dilute the monthly numbers), while Hunter's share of submitted decks is flat to slightly rising (31.6% to 33.0%). Players dropped the card and kept the class. A replicator-style test explains why a class-level model would find nothing: month-to-month archetype shares sit at their sampling noise floor (mean absolute change 2.9 pp against a 2.2 pp binomial standard error), whereas card adoption around shocks moves by 40–60 pp. Card adoption is the predictable target.

C. Prediction Pipeline

Real pre-expansion seeds evolving under the expanded pool adopt almost none of the cards the community adopted. A single-card swap changes a deck's win rate by about 1 pp, below the noise floor of affordable per-deck evaluation. At the budgets tested, unguided whole-deck evolution did not reliably discover the relevant card improvements, leaving new cards near the floor that mutation alone produces (Section V-D). A *marginal-value probe* measures a single card's worth directly, without evolving anything; *probe-biased evolution* is the ordinary evolutionary loop with the probe's values steering only the mutation step. The probe isolates the card variable directly: for each candidate card and class it swaps the card into a fixed slot of five real host decks, plays a fixed 60-game schedule against a fixed nine-deck field of real decks under the relevant balance (2,700 games per card and side), and records the mean win-rate delta. For the Naxxramas launch the probe's top Hunter cards are the community's staple package (Webspinner $+2.8$ pp, Mad Scientist $+2.3$ pp, Haunted Creeper $+2.2$ pp, Loatheb $+1.0$ pp, Sludge Belcher $+0.9$ pp). Its misses are cards whose value depends on the deck built around them: Duplicate probes at -5.5 pp in Mage against a real adoption of 43.4%, and Undertaker at -1.4 pp in Hunter against 20.6%. A fixed-slot swap into an ordinary host cannot express what a card is worth in a deck designed for it, a limit of the probe's design that Section VII returns to.

Probe-biased evolution then proposes probed-positive cards more often and replaces probed-negative slots more often, a stand-in for players' deliberate testing, with selection untouched (24 fixed games per evaluation, 25 generations). The bias strength was chosen by a sweep over four values on the expansion era at a reduced budget; the nerf era uses the same value without retuning. The mutation operator respects deck copy structure: real 2014 decks average 18 distinct cards in 30 slots, and mutation up-weights both doubling a held single copy and removing one, with legendaries capped at one. Raw inclusion counts overstate adoption: a card in all of a niche's 18–40 winning decks reads as exactly 100%, and the evolved decks carry a few more distinct cards than real decks do, which inflates every number. We correct this once, by rescaling the predicted adoption so its total matches what real pre-shock

TABLE IV
HEARTHSTONE NERF-RESPONSE PREDICTION (HUNTER, WITHIN-CLASS ADOPTION; THREE-SEED ENSEMBLE WITH SEED RANGE).

card	pre-nerf	predicted	seed range	real
Starving Buzzard (nerfed)	73.4%	6.1%	2–9%	16.8%
Leeroy Jenkins (nerfed)	34.8%	14.0%	4–31%	12.0%
Unleash the Hounds	84.4%	64.9%	24–95%	69.5%
Webspinner	54.4%	50.0%	43–54%	73.7%

decks carried; the rescaling keeps every card's order intact, so it fixes the levels without touching the rankings. Each experiment runs at three seeds and the reported forecast is their mean.

D. One-Shot Results

For the expansion, probe-biased evolution matches the real direction on 31 of the 38 Hunter cards that moved by at least 5 pp (19 expected by chance), with Hunter movers Spearman 0.78; across the three classes it calls 108 of 135 movers and the pooled adoption error is 5.5 pp. Unbiased evolution under the same configuration leaves the new cards at the mutation floor, Webspinner 2% against 54%, Mad Scientist 5% against 33%, Sludge Belcher 4% against 12%, with Hunter movers Spearman 0.64; the aggregate metrics move less than the cards do because the roughly 100 unmoved Classic cards dominate them. Mad Scientist is predicted at 19.1% within Hunter against a real 32.7%, Sludge Belcher at 16.7% against 11.6%, and Haunted Creeper and Webspinner rise to 36% and 34% against 58% and 54%. Kel'Thuzad, Unstable Ghoul, and Zombie Chow are correctly left near zero. The misses are the build-around cards the probe undervalues: Undertaker stays at 2% against 21%, Duplicate at 2% against 43%, and Death's Bite reaches 8% against 60%, with a seed range of 2–19%. Mad Scientist also shows how much the probe depends on its host decks. It is adopted in Hunter, whose real decks run the secrets it fetches, and stays at 3% in Mage against a real 41%, where the probe's Mage hosts carry no secrets and value the card below zero.

The nerf is the cleaner test, with no new cards and the real post-shock pool equal to the simulated pool. Table IV and Figure 2 show the headline cards. Both nerfed cards collapse to within a few points of their real post-nerf rates (Buzzard 6.1% predicted against 16.8% real, Leeroy 14.0% against 12.0%), with Hunter movers Spearman 0.73 and pooled adoption error 5.8 pp. Unleash the Hounds, untouched by the patch, is kept at 64.9% against 69.5%, though its seed range (24–95%) is the widest of any card. What the pipeline misses is the rises: Webspinner is held near flat where reality moved from 54% to 74%, and Mad Scientist stays at 29% where reality doubled to 64%. The removal probe carries the sign of the Buzzard response on its own: deleting Buzzard from patched-world Hunter decks gains $+2.5$ pp. It does not carry the Leeroy response. Removing Leeroy costs a strong pilot 1.7 pp, so the bias favours keeping the card, and the cut to 14% comes from selection under the patched balance.

The no-change baseline ties the prediction on the overall adoption Spearman after the expansion (0.71–0.77 against 0.72–0.78 by class) and beats it after the nerf (0.88–0.92 against 0.71–0.74), for the same reason it does in Pokémon: most cards do not move, and a baseline that copies the roughly 100 unmoved cards is hard to beat on a rank metric dominated by non-movers. The prediction's advantage is confined to the movers, in direction and magnitude.

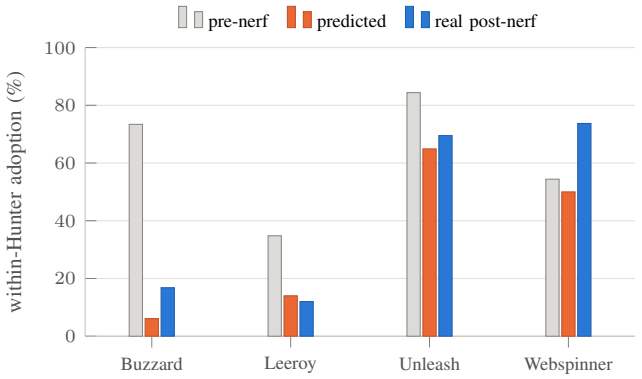


Fig. 2. Hearthstone nerf-response prediction against reality (three-seed ensemble). The nerfed pair falls to about where it really landed; Unleash the Hounds is kept near its real rate, and Webspinner's rise is missed. Gray gives the pre-nerf baseline, orange the model, blue the real outcome.

E. Rolling-Trajectory Results

A rolling variant predicts the whole post-shock trajectory at the archive's data resolution, five periods bounded by the two patch dates, and separates two seeding regimes. In *anchored* mode each period is seeded from the previous period's real decks; in *free-running* mode only the first period is real and every later period starts from the model's own previous elite population. Each mode runs at three seeds, ten generations per period, and the reported trajectory is their mean.

Anchored one-step prediction holds up in every period: movers Spearman runs 0.56–0.90 for every class and period (all $p \leq 0.03$), with per-period level error of 2.4–4.0 pp. The predicted Buzzard trajectory tracks the real collapse (Figure 3), 73%, 43%, 11%, and 6% against a real 79%, 53%, 16%, and 18%, and Mad Scientist and Webspinner sit within a few points of reality from the second period on. The free-running forecast also reaches the collapse, one period late (74% where reality had fallen to 53%, then 29% and 3%), and all three seeds drop the card by the final period. The cost of free-running falls on the cards that rose. With no real decks re-entering the loop, the evaluation gauntlet is drawn from the model's own population, which never demands the new cards, and their adoption lags reality by 20–45 pp through the whole trajectory: Mad Scientist reaches 34% against 60%, Webspinner 38% against 68%. Rank agreement on movers falls to 0.30–0.67 after the first period and loses significance for Mage after the nerf, and level error rises to 3.5–5.5 pp. Re-anchoring on real decks each period removes the lag. The useful free-running horizon is one period, roughly six weeks; beyond it the forecast keeps the direction of the shock and loses its levels.

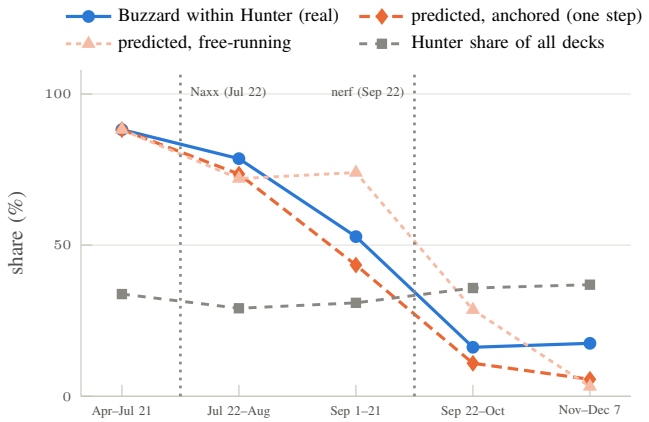


Fig. 3. The Hearthstone Buzzard trajectory at the archive's data resolution, with both rolling modes (three-seed means). All model series share the first point as input; everything after it is prediction. Anchored, re-seeded from real decks, tracks the collapse; free-running, re-seeded from its own population, reaches it one period late. Hunter's class share (gray) never leaves its noise band.

F. The Hidden-Information Ablation

Every result above uses agents that cannot see the opponent's hand. Repeating the probe and the expansion-era evolution with agents trained and run with both hands visible measures what hidden information contributes. At the probe level the two protocols agree with each other more than either agrees with the population. Per-class rank agreement with real adoption is 0.35 against 0.34 for Hunter, 0.31 against 0.38 for Mage, and 0.59 against 0.61 for Warrior (hidden information first; only the Warrior values are significant, $p < 0.005$). Single cards move more than the aggregates: Mad Scientist in Hunter probes at $+2.3$ pp hidden against $+3.3$ pp visible, and Duplicate at -5.5 pp against -4.0 pp.

At the evolution level, three seeds per protocol under the same configuration, hidden information gives a lower pooled adoption error (5.5 against 5.8 pp) and a higher mover direction rate (0.80 against 0.75), a modest aggregate gain. The card-level differences are larger. With visible hands the pipeline over-adopts Mad Scientist (42.8% against a real 32.7%, one seed at 76%): an opponent who can see the hand plays around nothing, so the card's body is pure tempo. It leaves Death's Bite at 1.6% (8.1% under hidden information) and Webspinner at 13.4% (33.9%). Which cards evolution finds depends on the information its pilots have, and the aggregate metrics understate the difference.

VI. Cross-Game Synthesis

The two studies used different games, shocks, codebases, and ground truth. They agree on four findings, and on the evaluation practice the findings recommend.

A. Composition versus Movement

In both games the no-change baseline wins composition: Pokémon no-change beats the framework on overlap in all four grid cells, and the Hearthstone no-change baseline ties the pipeline on the overall adoption rank correlation after the expansion and beats it after the nerf. In both, overlap sits near its ceiling because a metagame is mostly stable across a shock, so it scores the fraction that did not move. What each pipeline does predict is the movement: Pokémon edit-distance delta beats no-change 42–0 at the high-ladder cutoff, and Hearthstone probe-biased evolution predicts adoption direction and magnitude on the cards that moved. A test a do-nothing baseline can pass is measuring how stable the meta is, not how good the method is. What is left, how much the meta moves and which way, is where a real prediction shows itself, and both studies reach it independently: one by replicating an existing framework, one by building a new pipeline.

B. Self-Referential Loops

Each pipeline contains a place where the model's own output feeds its next input, and in each, that loop degrades prediction. The Pokémon usage refresh rebuilds teams from pickrates the builder itself wrote, and the Hearthstone free-running forecast draws its evaluation gauntlet from its own previous elite population. Neither has an outcome signal that pulls it back toward what players did: Pokémon predictions worsen when the refresh is faithful, and the Hearthstone forecast, evaluated against opponents drawn from its own population, adopts the new cards at about half their real rate and calls the nerf one period late. Both are fixed the same way, by pulling the method back to real data at each step: keeping the real month's statistics rather than the simulation's, or re-seeding each period from real decks. That the same failure and the same fix appear in two unrelated systems suggests the loop is a property of population-feedback prediction itself, the same drift that population-based self-play has to be designed around [53], [54].

C. Use of Simulated Outcomes

In the tested ABC-Meta configuration, the team builder scores species by their observed usage times their base stat total. Predicted usage comes from counting appearances on the resulting teams. A species on a losing team counts just as much as one on a winning team, so the forecast can be computed without playing the battles. The battle-free rerun produces almost unchanged predictions.

Hearthstone uses the games at two stages. The probe inserts a card into fixed real decks and measures the change in win rate. Evolution proposes cards with positive probe values more often, then plays the resulting decks to decide which to retain. At the budgets tested, evolution without the probe left new cards near the mutation floor. Selection still matters: the probe favours keeping Leeroy, but its adoption falls during evolution under the patched rules.

Our Pokémon surrogate is trained on simulated battle outcomes. During search, it estimates a candidate team's win probability against reference teams sampled from real usage, replacing fresh battles for each comparison. Counting species in the teams found by search produces too much change, so the final forecast blends the model's output with the previous month's usage. Replacing that output with random noise and applying the same blend does not reproduce the

result. This control supports a contribution from the learned model, although it does not establish reliable predictions of which individual species will rise or fall.

D. Direction Prediction

Neither study can say which particular unit will rise or fall. In Pokémon, the framework calls mover direction right about half the time (0.53), and the small positive rank correlations it does show survive a control that carries no real information about the change: shuffle a ranking and watch it drift back toward the middle, and you get correlations of the same size for nothing. In Hearthstone the probe's rank agreement with real adoption reaches significance only for Warrior; at 22 cards per class, Hunter and Mage stay below 0.4 under either information protocol. Calling the direction of specific movement is the problem this pair of studies leaves open.

E. Evaluation Recommendations

Both studies argue for the same evaluation practice. Monthly Pokémon usage transitions and period-resolved Hearthstone adoption both supply many objective test cases on a fixed schedule, where hand-curated ban scenarios cap evaluation at a handful. In both, a no-change baseline is mandatory, paired significance tests should accompany point estimates, and the reported metrics should be ones the baseline cannot trivially satisfy. The Pokémon study adds the rating cutoff as a free parameter that flips the headline comparison, and the Hearthstone study shows that how you read a predicted level off the model matters as much as the model itself. Evaluating across regimes and cutoffs costs little; every boundary condition reported in this paper was found that way.

VII. Limitations

The Pokémon drift evaluation covers one tier of one generation; gen9ou is the most played tier, but tier-specific dynamics could shift the numbers. The exact-scenario rerun spans four tiers of Generation 8 only, with fidelity notes on the reconstructions that current validators make impossible. The 10,000-battle drift budget adds pickrate sampling noise a larger budget would reduce, though Section IV-E bounds how much it can matter. The surrogate's strongest configuration carries a post-hoc read-out-family caveat, partly answered by the cross-cutoff transfer but not by a prospective test on future months. The rating-0 arm ran the static configuration only.

The Hearthstone study uses three of the nine classes, so cross-class effects are invisible to it. The probe swaps a card into fixed slots of ordinary host decks, so cards whose value comes from a deck built around them (Duplicate, Undertaker) are undervalued by design, and Death's Bite is adopted well below its real rate at the 200-iteration search budget used throughout. The one-shot predictions are three-seed ensembles with the spread reported, but individual cards still swing across seeds. The probe aggregates five hosts and 60 games per card but is itself a single run. Across both studies, the ground-truth archives are community-maintained and carry the collection biases of the populations that produced them.

VIII. Conclusion

We asked what simulation predicts about a shifting metagame, and answered it in two games with two independent pipelines. In Pokémon Showdown we replicated a published meta discovery framework, reproduced its four ban scenarios within one species, and mapped its operating envelope on a completed grid of generations and rating cutoffs. In Hearthstone we built a marginal-value probe and probe-biased evolution and predicted card adoption across an expansion and a nerf against 33,697 real decklists. The two agree. A no-change baseline wins the composition metric in both, which saturates near its ceiling and measures stability rather than prediction; what a simulator recovers is how much the metagame moves, on the units that moved. Both pipelines contain a loop that feeds on their own output and degrades until grounded in real data, and in both the real starting usage carries the predictive content, with simulation only adding precision on top. The direction of specific movement remains unpredicted by every method we tested. We contribute two methods aimed at the movement signal, a fast learned stand-in for simulation, read out by blending toward no change, for Pokémon and a marginal-value probe for Hearthstone, and a shared evaluation standard: a no-change baseline reported alongside every result, metrics that baseline cannot trivially satisfy, and more than one regime and cutoff.

Acknowledgments

Claude models (Anthropic), primarily Claude Fable 5, were used throughout this work: as a coding assistant for both experimental pipelines, to run and monitor the experiments, and to draft and revise the manuscript under the authors' direction. The authors reviewed all results and text and take full responsibility for them.

References

[1] A. Saravanan and M. Guzdial, "A framework for predicting the impact of game balance changes through meta discovery," *IEEE Transactions on Games*, vol. 16, no. 4, pp. 821–830, 2024.

[2] A. Jaffe, A. Miller, E. Andersen, Y.-E. Liu, A. Karlin, and Z. Popović, "Evaluating competitive game balance with restricted play," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 8, no. 1, 2012, pp. 26–31.

[3] V. Volz, G. Rudolph, and B. Naujoks, "Demonstrating the feasibility of automatic game balancing," in *Proceedings of the Genetic and Evolutionary Computation Conference 2016*, 2016, pp. 269–276.

[4] N. Tomašev, U. Paquet, D. Hassabis, and V. Kramnik, "Assessing game balance with AlphaZero: Exploring alternative rule sets in chess," *arXiv preprint arXiv:2009.04374*, 2020.

[5] F. de Mesentier Silva, R. Canaan, S. Lee, M. C. Fontaine, J. Togelius, and A. K. Hoover, "Evolving the hearthstone meta," in *2019 IEEE Conference on Games (CoG)*, 2019.

[6] D. Hernandez, C. T. T. Gbadamosi, J. Goodman, and J. A. Walker, "Metagame autobalancing for competitive multiplayer games," in *2020 IEEE Conference on Games (CoG)*, 2020.

[7] W. Kavanagh, A. Miller, G. Norman, and O. Andrei, "Balancing turn-based games with chained strategy generation," *IEEE Transactions on Games*, vol. 13, no. 2, pp. 113–122, 2021.

[8] J. Pfau, A. Liapis, G. N. Yannakakis, and R. Malaka, "Dungeons & replicants II: Automated game balancing across multiple difficulty dimensions via deep player behavior modeling," *IEEE Transactions on Games*, vol. 15, no. 2, pp. 217–227, 2023.

[9] H. Yue, H. Liu, and J. Chen, "A gospel for MOBA game: Ranking-preserved hero change prediction in Dota 2," *IEEE Transactions on Games*, vol. 14, no. 2, pp. 191–201, 2022.

[10] Y. He, C. Tran, J. Jiang, K. Burghardt, E. Ferrara, E. Zheleva, and K. Lerman, "Heterogeneous effects of software patches in a multiplayer online battle arena game," in *Proceedings of the 16th International Conference on the Foundations of Digital Games*, 2021.

[11] C.-S. Lee and I. Ramler, "Investigating the impact of game features and content on champion usage in League of Legends," in *Proceedings of the 10th International Conference on the Foundations of Digital Games*, 2015.

[12] ——, "Identifying and evaluating successful non-meta strategies in League of Legends," in *Proceedings of the 12th International Conference on the Foundations of Digital Games*, 2017.

[13] N. Elbert, E. Paskovski, N. Stein, and C. M. Flath, "The layers of change: Multilevel adaptations to exogenous game patches in CS:GO," in *2023 IEEE Conference on Games (CoG)*, 2023, pp. 1–4.

[14] A. Pedrassoli Chitayat, F. Block, J. Walker, and A. Drachen, "Beyond the meta: Leveraging game design parameters for patch-agnostic esport analytics," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 19, no. 1, 2023, pp. 116–125.

[15] H. Heo and H. K. Kim, "When to switch, not just what: Transition quality prediction in Clash Royale," 2026, arXiv:2605.21868.

[16] J. Codsi and A. Vetta, "A case study in learning in metagames: Super smash bros. melee," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 17, no. 1, 2021, pp. 2–9.

[17] G. Luo and Pokémon Showdown contributors, "Pokémon showdown," https://github.com/smogon/pokemon-showdown.

[18] S. Lee and J. Togelius, "Showdown AI competition," in *2017 IEEE Conference on Computational Intelligence and Games (CIG)*, 2017.

[19] D. Huang and S. Lee, "A self-play policy optimization approach to battling Pokémon," in *2019 IEEE Conference on Games (CoG)*, 2019, pp. 1–4.

[20] S. Karten, J. Grigsby, T. Upaa, J. Bae, S. Hong, H. Jeong, J. Jung, K. Kerdthaisong *et al.*, "The PokeAgent challenge: Competitive and long-context learning at scale," 2026, arXiv:2603.15563; NeurIPS 2025 Competition Track.

[21] H. Sahovic, "poke-env: A python interface for training reinforcement learning pokémon bots," https://github.com/hsahovic/poke-env, 2019.

[22] J. Grigsby, Y. Xie, J. Sasek, S. Zheng, and Y. Zhu, "Human-level competitive Pokémon via scalable offline reinforcement learning with transformers," in *Reinforcement Learning Conference*, 2025, arXiv:2504.04395.

[23] S. Reis, L. P. Reis, and N. Lau, "VGC AI competition – a new model of meta-game balance AI competition," in *2021 IEEE Conference on Games (CoG)*, 2021, pp. 1–8.

[24] S. Reis, A. L. Martins, R. Novais, and F. Alves, "A new rules balance track for the Pokémon VGC AI competition 2.0," in *2025 IEEE Conference on Games (CoG)*, 2025, pp. 1–8.

[25] S. Reis, R. Novais, L. P. Reis, and N. Lau, "An adversarial approach for automated Pokémon team building and metagame balance," *IEEE Transactions on Games*, vol. 16, no. 2, pp. 365–375, 2024.

[26] A. F. Ramos and T. Soria, "From rules to Nash equilibria: A Lean 4 case study in game-theoretic analysis of a competitive trading card game," 2026, arXiv:2607.08692.

[27] P. García-Sánchez, A. Tonda, G. Squillero, A. M. Mora, and J. J. Merelo, "Evolutionary deckbuilding in Hearthstone," in *2016 IEEE Conference on Computational Intelligence and Games (CIG)*, 2016.

[28] A. Bhatt, S. Lee, F. de Mesentier Silva, C. W. Watson, J. Togelius, and A. K. Hoover, "Exploring the Hearthstone deck space," in *Proceedings of the 13th International Conference on the Foundations of Digital Games*, 2018.

[29] M. C. Fontaine, S. Lee, L. B. Soros, F. de Mesentier Silva, J. Togelius, and A. K. Hoover, "Mapping Hearthstone deck spaces through MAP-Elites with sliding boundaries," in *Proc. Genetic and Evolutionary Computation Conference (GECCO)*, 2019.

[30] Y. Zhang, M. C. Fontaine, A. K. Hoover, and S. Nikolaidis, "Deep surrogate assisted MAP-Elites for automated Hearthstone deckbuilding," in *Proceedings of the Genetic and Evolutionary Computation Conference*, 2022, pp. 158–167.

[31] C. França, Z. Zhou *et al.*, "CreativeStone: A creativity booster for Hearthstone card decks," *IEEE Transactions on Games*, vol. 16, no. 1, pp. 214–224, 2024.

[32] Z. Chen, C. Amato, T.-H. D. Nguyen, S. Cooper, Y. Sun, and M. Seif El-Nasr, "Q-DeckRec: A fast deck recommendation system for collectible card games," in *2018 IEEE Conference on Computational Intelligence and Games (CIG)*, 2018, pp. 1–8.

[33] T. Chen and S. Guy, "Chaos cards: Creating novel digital card games through grammatical content generation and meta-based card evaluation," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 16, no. 1, 2020, pp. 196–202.

[34] G. E. M. Long, D. Pérez-Liébana, and S. Samothrakis, "Balancing wargames through predicting unit point costs," in *2023 IEEE Conference on Games (CoG)*, 2023, pp. 1–8.

[35] A. Dockhorn and S. Mostaghim, "Introducing the Hearthstone-AI competition," *arXiv preprint arXiv:1906.04238*, 2019.

[36] M. Świechowski, T. Tajmajer, and A. Janusz, "Improving Hearthstone AI by combining MCTS and supervised learning algorithms," in *2018 IEEE Conference on Computational Intelligence and Games (CIG)*, 2018.

[37] N. Justesen, T. Mahlmann, S. Risi, and J. Togelius, "Playing multiaction adversarial games: Online evolutionary planning versus tree search," *IEEE Transactions on Games*, vol. 10, no. 3, pp. 281–291, 2018.

[38] J. R. Long, N. R. Sturtevant, M. Buro, and T. Furtak, "Understanding the success of perfect information Monte Carlo sampling in game tree search," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 24, no. 1, 2010, pp. 134–140.

[39] A. Dockhorn, M. Frick, Ü. Akkaya, and R. Kruse, "Predicting opponent moves for improving Hearthstone AI," in *Proc. Int. Conf. Information Processing and Management of Uncertainty (IPMU)*. Springer, 2018, pp. 621–632.

[40] Vicious Syndicate, "[Classic] vS Data Reaper Report #1," https://www.vicioussyndicate.com/classic-vs-data-reaper-report-1/, Apr. 2021, accessed 2026-09-09.

[41] Smogon, "Smogon usage statistics," https://www.smogon.com/stats/.

[42] P. García-Sánchez, A. Tonda, A. Fernández-Ares, and A. M. Mora, "Hearthstone decks dataset," 2023, doi:10.5281/zenodo.10198504.

[43] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," *arXiv preprint arXiv:1504.04909*, 2015.

[44] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, "Deep sets," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[45] A. V. Kokkinakis, P. York, M. S. Patra, J. Robertson, B. Kirman, A. Coates, A. P. Chitayat, S. Demediuk, A. Drachen, J. Hook, I. Nolle, O. Olarewaju, D. Slawson, M. Ursu, and F. Block, "Metagaming and metagames in esports," *International Journal of Esports*, vol. 1, no. 1, 2021, https://www.ijesports.org/article/51, accessed 2026-09-09.

[46] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021, section 5.2, https://otexts.com/fpp3/.

[47] pkmn contributors, "@pkmn/sim: Pokémon showdown as a library," https://github.com/pkmn/ps.

[48] Y. Jin, M. Olhofer, and B. Sendhoff, "A framework for evolutionary optimization with approximate fitness functions," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 5, pp. 481–494, 2002.

[49] C. D. Rosin, "Multi-armed bandits with episode context," *Annals of Mathematics and Artificial Intelligence*, vol. 61, pp. 203–230, 2011.

[50] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, Y. Chen, T. Lillicrap, F. Hui, L. Sifre, G. van den Driessche, T. Graepel, and D. Hassabis, "Mastering the game of Go without human knowledge," *Nature*, vol. 550, pp. 354–359, 2017.

[51] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis, "A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play," *Science*, vol. 362, no. 6419, pp. 1140–1144, 2018.

[52] A. M. Mora, A. Tonda, A. J. Fernández-Ares, and P. García-Sánchez, "Looking for archetypes: Applying game data mining to Hearthstone decks," *Entertainment Computing*, vol. 43, p. 100498, 2022.

[53] M. Lanctot, V. Zambaldi, A. Gruslys, A. Lazaridou, K. Tuyls, J. Pérolat, D. Silver, and T. Graepel, "A unified game-theoretic approach to multiagent reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[54] D. Balduzzi, M. Garnelo, Y. Bachrach, W. M. Czarnecki, J. Pérolat, M. Jaderberg, and T. Graepel, "Open-ended learning in symmetric zero-sum games," in *Proceedings of the 36th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 434–443.